

BUILDING A SOC OVERVIEW DASHBOARD

Student Activity Guide — Point-and-Click Only, No XML Editing

This guide builds one complete, filterable, professional-looking SOC dashboard on the BOTS v2 dataset entirely through Splunk's visual dashboard editor. Every step uses buttons, dropdowns, and text boxes you can see on screen — you will not open the Source/XML view at any point. Follow every part in order; each one depends on the last.

Prerequisites

- Splunk Enterprise installed with the BOTS v2 Attack-Only dataset loaded (index=botsv2), per Lesson 2.
- Comfort writing basic SPL: index, stats, sort, table (Lessons 4–6).
- Comfort using fieldsummary to discover real field names (Lesson 7) — you will need this if your environment's field names differ from the ones used in this guide.

A note before you start: every search in this guide uses fields that exist in almost every Splunk environment — sourcetype, host, _time. Later, when we add detection panels that reference usernames, we will use a technique that survives field names being different in your data. Keep Lesson 7's fieldsummary trick in your back pocket in case any panel comes back empty.

PART 1 Create the Dashboard Shell

GOAL: An empty, saved, titled dashboard you can add panels to.

1. In Splunk, go to Search & Reporting.
2. Click the Dashboards tab in the top navigation bar.
3. Click Create New Dashboard.
4. Title: BOTS v2 Security Overview
5. Description: An Overview for Security Alerts in BotsV2
6. Dashboard Type: choose Classic Dashboard (not Dashboard Studio).
7. Click Create. You will land in an empty dashboard editor.

If you ever see a screen asking you to choose between a grid layout and an absolute/free-form layout, choose Grid. Everything in this guide assumes the grid layout.

PART 2 Build the Total Events Panel

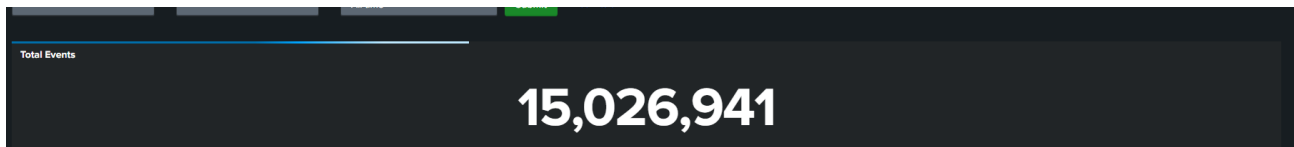
GOAL: One big number showing total events in the dataset — your headline KPI tile.

1. Click + Add Panel (top toolbar).
2. Choose New, then Single Value under Visualization types.
3. In the Search box that appears, type or paste the search below.

```
index=botsv2 earliest=0  
| stats count
```

4. Panel Title: Total Events
5. Click Add to Dashboard.

You should now see one panel with a large number — your total event count.



PART 3 Build the Three Breakdown Panels

GOAL: Three pie charts showing the composition of your data by sourcetype, by host, and by hour of day.

Repeat the same four clicks for each of the three panels below: + Add Panel → New → Pie Chart → paste the search → title it → Add to Dashboard.

Panel A — Events by Sourcetype

```
index=botsv2 earliest=0  
| stats count by sourcetype  
| sort - count  
| head 10
```

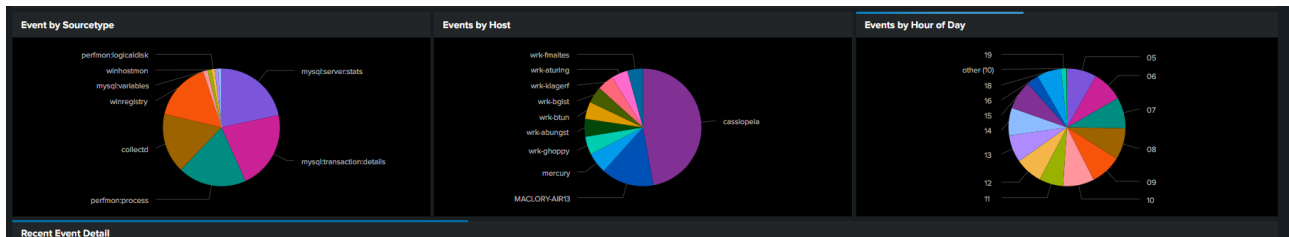
Panel B — Events by Host

```
index=botsv2 earliest=0  
| stats count by host  
| sort - count  
| head 10
```

Panel C — Events by Hour of Day

```
index=botsv2 earliest=0
| eval hour=strftime(_time,"%H")
| stats count by hour
| sort hour
```

Why these three fields specifically? sourcetype, host, and _time exist in every single event Splunk indexes, regardless of log source. Building your first version of any dashboard on fields you know exist — rather than fields you're hoping exist — is a deliberate habit worth keeping for the rest of your career.



PART 4 Build the Recent Event Detail Table

GOAL: A row-by-row table an analyst can actually read and drill into.

1. Click + Add Panel → New → Statistics Table.
2. Paste the search below.

```
index=botsv2 earliest=0
| table _time host source sourcetype
| sort - _time
| head 100
```

3. Panel Title: Recent Event Detail
4. Click Add to Dashboard.

Recent Event Detail			
_time	host	source	sourcetype
2017-08-31 11:17:11.384	jabbah	stream:http	stream:http
2017-08-31 11:17:00.160	jabbah	stream:http	stream:http
2017-08-31 11:17:00.159	jabbah	stream:http	stream:http
2017-08-31 11:17:00.159	jabbah	stream:http	stream:http
2017-08-31 11:17:00.045	jabbah	stream:http	stream:http
2017-08-31 11:17:00.041	jabbah	stream:http	stream:http
2017-08-31 11:17:00.040	jabbah	stream:http	stream:http
2017-08-31 11:16:59.990	jabbah	stream:http	stream:http
2017-08-31 11:16:59.990	jabbah	stream:http	stream:http
2017-08-31 11:16:59.988	jabbah	stream:http	stream:http

PART 5 Arrange the Layout

GOAL: A clean grid: KPI on top, three pies in a row, table across the bottom.

Drag each panel by its title bar to reposition it, and drag its bottom-right corner to resize it, until your dashboard reads top to bottom as:

```
[ Total Events ..... full width ]
[ By Sourcetype ] [ By Host ] [ By Hour of Day ]
[ Recent Event Detail ..... full width ]
```

1. Click Save (top right) once the layout matches.



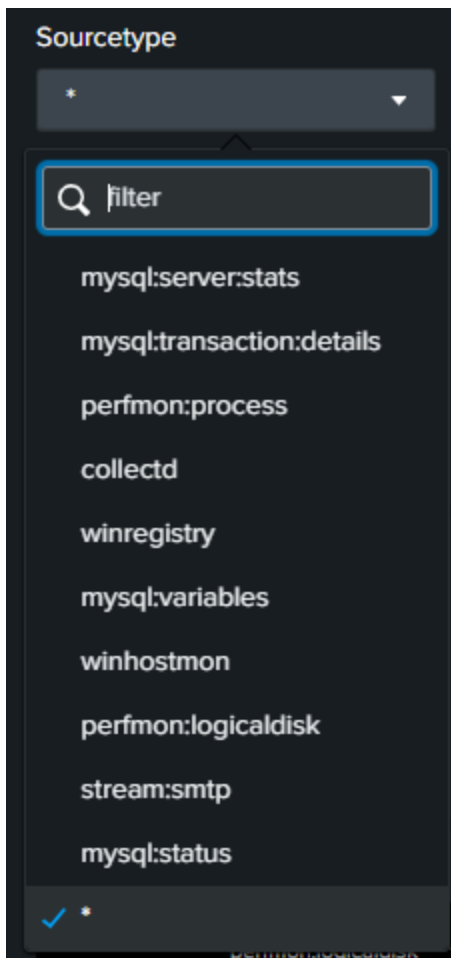
PART 6 Add the Three Filter Inputs

GOAL: Sourcetype, Host, and Time Range controls at the top of the dashboard, matching a professional SIEM's filter bar.

Input 1 — Sourcetype dropdown

1. Click Edit (top right) to enter edit mode, if not already there.
2. Click + Add Input, then choose Dropdown.
3. A new input box appears. Click its pencil (edit) icon.
4. Label: Sourcetype
5. Token: st_tok
6. Under Static Options / Search, choose Search, then paste:

```
index=botsv2 earliest=0 | stats count by sourcetype | sort - count
```
7. Field For Label: sourcetype
8. Field For Value: sourcetype
9. Default: type * (a single asterisk) — this makes “all sourcetypes” the starting state.
10. Click Apply / Save on the input.



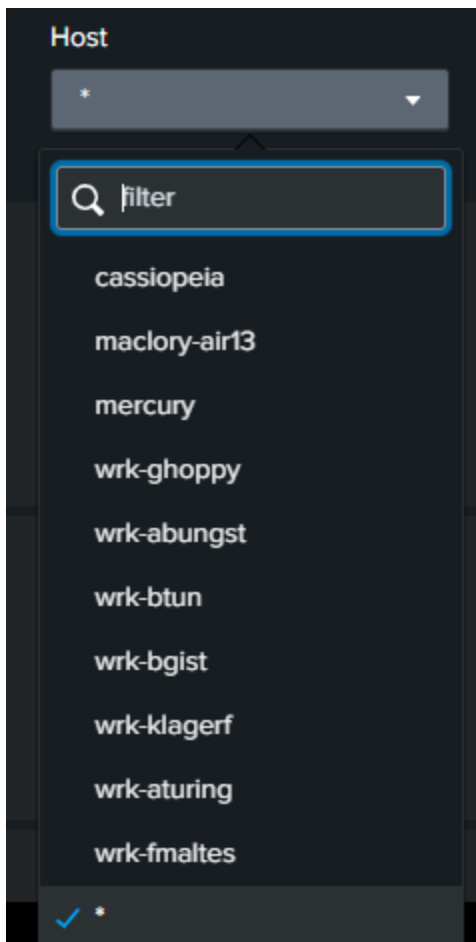
Input 2 — Host dropdown

Repeat the exact same steps as Input 1, with these values instead:

1. Label: Host
2. Token: host_tok
3. Search:

```
index=botsv2 earliest=0  
| stats count by host  
| sort - count  
| head 10
```

4. Field For Label: host
5. Field For Value: host
6. Default: *



Input 3 — Time Range picker

1. Click + Add Input again, but this time choose Time (not Dropdown).
2. Label: Time Range
3. Token: tr
4. Default: set Earliest to 0 and leave Latest blank (meaning “now”).

Alerts in BotsV2

Host Time Range [Hide Filters](#)

Presets

Real-time	Relative	Other	
30 second window	Today	Last 15 minutes	All time
1 minute window	Week to date	Last 60 minutes	
5 minute window	Business week to date	Last 4 hours	
30 minute window	Month to date	Last 24 hours	
1 hour window	Year to date	Last 7 days	
All time (real-time)	Yesterday	Last 30 days	
	Previous week		
	Previous business week		
	Previous month		
	Previous year		

> Relative

> Real-time

> Date Range

> Date & Time Range

> Advanced

Common mistake: adding "Time Range" as a Dropdown instead of the dedicated Time input type. A Dropdown time control has nothing to select from unless you manually build a list of choices, and it will look permanently greyed out. Always use the Time input type for time ranges.

PART 7 Turn On the Submit Button

GOAL: A visible green Submit button, so panels only refresh when you choose — not on every single click.

1. While still in Edit mode, look at the top right area near your inputs for a checkbox labeled Autorun Dashboard.
2. Uncheck it.
3. Click Save.

Exit edit mode. A Submit button should now appear next to your three filters. Changing a dropdown should now do nothing until you click Submit — that is the correct, intended behavior.

[Hide Filters](#)

PART 8 Connect the Filters to Every Panel

GOAL: Every panel actually responds when you change a filter and click Submit, instead of the inputs sitting there disconnected.

Adding inputs creates the tokens (\$st_tok\$, \$host_tok\$, \$tr\$), but no panel uses them until you edit each panel's search to include them. For each of the five panels: click Edit, click the panel, click the magnifying-glass / Edit Search icon, and replace the search with the version below.

Total Events

```
index=botsv2 sourcetype=$st_tok|s$ host=$host_tok|s$  
| stats count
```

Edit Search

Title

Total Events

Search String

index=botsv2 sourcetype=\$st_tok|s\$ host=\$host_tok|s\$
| stats count

Run Search

Time Range

Shared Time Picker (tr) ▾

Auto Refresh Delay ?

No auto refresh ▾

Refresh Indicator

Progress bar ▾

Cancel

Convert to Report

Apply

Events by Sourcetype (filter by Host only)

```
index=botsv2 host=$host_tok|s$  
| stats count by sourcetype  
| sort - count  
| head 10
```

General

Token Options

Static Options

Dynamic Options

Content Type

Search String

```
index=botsv2 earliest=0
host=$host_tok|s$ |
stats count by
sourcetype | sort -
count | head 10
```

Run Search

All time

Field For Label ?

sourcetype

Field For Value ?

sourcetype

Cancel

Apply

Don't filter this panel by Sourcetype — doing so would collapse its own breakdown to a single slice and defeat the point of the chart.

Events by Host (filter by Sourcetype only)

```
index=botsv2 sourcetype=$st_tok|s$
| stats count by host
| sort - count
| head 10
```

✕

✕

✕

Host

Time Range

T Text

⦿ Radio

▼ Dropdown

☑ Checkbox

▼ Multiselect

🔗 Link List

⦿ Time

General

Token Options

Static Options

+

Dynamic Options

Content Type

🔍

📄

Search String

```
index=botsv2 earliest=0
sourcetype=$st_tok|s$ |
stats count by host |
sort - count | head 10
```

Run Search 🔗

All time

▼

Field For Label ?

host

Field For Value ?

host

Cancel

Apply

Events by Hour of Day

```
index=botsv2 sourcetype=$st_tok|s$ host=$host_tok|s$
| eval hour=strftime(_time,"%H")
| stats count by hour
| sort hour
```

Edit Search

×

Title

Events by Hour of Day

Search String

```
index=botsv2 earliest=0 sourcetype=$st_tok|s$ host=$host_tok|s$ | eval hour
=strftime(_time,"%H") | stats count by hour | sort hour
```

Run Search

🔗

Time Range

Shared Time Picker (tr) ▼

Auto Refresh Delay ?

No auto refresh ▼

Refresh Indicator

Progress bar ▼

Cancel

Convert to Report

Apply

Recent Event Detail

```
index=botsv2 sourcetype=$st_tok|s$ host=$host_tok|s$
| table _time host source sourcetype
| sort - _time
| head 100
```

Edit Search

×

Title

Recent Event Detail

Search String

```
index=botsv2 earliest=0 sourcetype=$st_tok|s$ host=$host_tok|s$ | table _time
host source sourcetype | sort - _time | head 100
```

Run Search

🔗

Time Range

Shared Time Picker (tr) ▼

Auto Refresh Delay ?

No auto refresh ▼

Refresh Indicator

Progress bar ▼

Cancel

Convert to Report

Apply

Last step for every panel — the time range

1. Still editing that same panel, find its time range control (usually a clock icon next to the search box).

2. Change it from Custom / All Time to Shared Time Picker (this makes the panel obey your Time Range input instead of running its own fixed range).
3. Click Apply. Repeat for all five panels.

Notice earliest=0 is gone from inside every search above, compared to Parts 2–4. That's intentional: an earliest= typed inside a search always overrides the Time Range input around it. Leaving it in would silently ignore anything a user picks, forever.

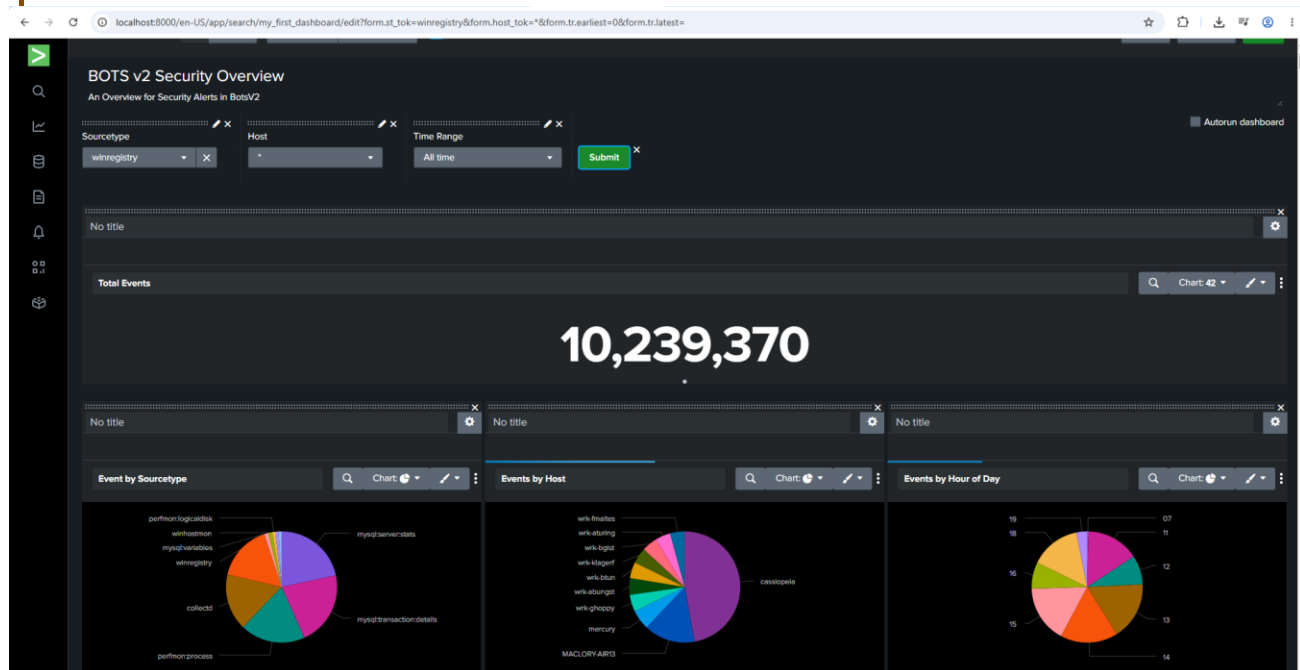
1. Click Save on the whole dashboard once all five panels are updated.

PART 9 Test It

GOAL: Proof the dashboard is genuinely wired, not just coincidentally showing correct data.

1. Exit edit mode.
2. Pick one specific Sourcetype from the dropdown (not “all” / blank).
3. Click Submit.
4. Confirm: Total Events drops to a smaller number, the Events by Sourcetype pie shrinks toward one slice, and Recent Event Detail only shows rows from that sourcetype.

If nothing changes when you do this, the most common cause is a panel whose search still says earliest=0 instead of using the Shared Time Picker, or a search that was never actually updated with the \$st_tok\$ / \$host_tok\$ tokens. Re-check Part 8 for that specific panel.



PART 10 Add a Detections Row

GOAL: Three new panels surfacing potential attack patterns, using techniques from Labs 3, 8, and 9.

A text header panel (optional, purely visual)

1. Click + Add Panel → New → look for a Text / Markdown / HTML panel type in your Splunk version's panel list.
2. Enter a heading such as: Potential Attacks & Security Alerts
3. Add to Dashboard, and drag it above the three panels below.

Detection 1 — High-Volume Accounts

This uses `coalesce()`, a function that checks several possible field names and uses whichever one actually has a value — the permanent fix for the exact field-name problem you ran into earlier in this course (user vs. Account_Name).

```
index=botsv2 sourcetype=$st_tok|s$ host=$host_tok|s$
| eval account=coalesce(user, Account_Name, user_name, src_user)
| where isnotnull(account)
| stats count as events dc(host) as hosts by account
| sort - events
| head 10
```

Edit Search

Title

High-Volume Accounts

Search String

```
index=botsv2 sourcetype=$st_tok|s$ host=$host_tok|s$
| eval account=coalesce(user, Account_Name, user_name, src_user)
| where isnotnull(account)
| stats count as events dc(host) as hosts by account
| sort - events
| head 10
```

Run Search

Time Range

Shared Time Picker (tr) ▾

Auto Refresh Delay ?

No auto refresh ▾

Refresh Indicator

Progress bar ▾

Cancel

Convert to Report

Apply

No title			⚙
High-Volume Accounts			🔍 Chart: 📊 ✎ ⋮
account ↕ ✎	events ↕ ✎	hosts ↕ ✎	
service3	24348	4	
-	9381	11	
root	3075	5	
MERCURY\$	2178	1	
WRK-ATURING\$	1769	2	
VENUS\$	1693	2	
WRK-BGIST\$	1569	2	
WRK-FMALTES\$	1551	2	
WRK-GHOPPY\$	1544	2	
WRK-BTUN\$	1538	2	

Add Panel → New → Statistics Table → paste the search → title it “High-Volume Accounts” → set its time range to Shared Time Picker, same as Part 8.

Detection 2 — Possible Network Scanning

```

index=botsv2 host=$host_tok|s$
| stats dc(dest_port) as unique_ports count as events by src_ip
| where unique_ports > 5
| sort - unique_ports
| head 10

```

Edit Search



Title Possible Network Scanning

Search String

```
index=botsv2 host=$host_tok|s$
| stats dc(dest_port) as unique_ports count as events by src_ip
| where unique_ports > 5
| sort - unique_ports
| head 10|
```

[Run Search](#)

Time Range

Shared Time Picker (tr) ▼

Auto Refresh Delay ?

No auto refresh ▼

Refresh Indicator

Progress bar ▼

Cancel

Convert to Report

Apply

No title			⚙
Possible Network Scanning			🔍 Chart:
src_ip ↕	unique_ports ↕	events ↕	
10.0.1.120	5769	26536	
160.153.91.7	725	2171	
10.0.1.100	190	215996	
45.77.65.211	129	18811	
67.132.183.74	97	175	
52.42.208.228	84	32777	
172.31.0.2	46	4925	
172.217.11.238	44	186	
52.40.10.231	39	80332	
67.132.183.9	39	97	

Add Panel → New → Statistics Table → title it “Possible Network Scanning” → Shared Time Picker.

Detection 3 — Web Path Enumeration

```
index=botsv2 sourcetype="stream:http" host=$host_tok|s$ status=404
| stats count as attempts dc(uri) as unique_paths by src_ip
| where unique_paths > 3
| sort - attempts
| head 10
```

Edit Search



Title **Web Path Enumeration**

Search String

```
index=botsv2 sourcetype="stream:http" host=$host_tok|s$ status=404  
| stats count as attempts dc(uri) as unique_paths by src_ip  
| where unique_paths > 3  
| sort - attempts  
| head 10|
```

[Run Search](#)

Time Range

Shared Time Picker (tr) ▼

Auto Refresh Delay ?

No auto refresh ▼

Refresh Indicator

Progress bar ▼

Cancel

Convert to Report

Apply

Web Path Enumeration



Chart:



i	Time	Event
>	8/31/17 11:17:11.304 AM	{ [-] accept: /*/* accept_language: en-US,en;q=0.8 bytes: 3679 bytes_in: 1523 bytes_out: 2156 c_ip: 212.83.203.115 cookie: store=default; mage-translation-storage=%7B%7D; mage-translation-file-version=%7B%7D; PHPSESSID=io259vpm8q9hbsjcjalum14ms3; form_key=P5QjF09iujN41DsK; mage-cache-storage=%7B%7D; mage-cache-storage-section-invalidation=%7B%7D; recently_viewed_product=%7B%7D; recently_viewed_product_previous=%7B%7D; recently_compared_product=%7B%7D; recently_compared_product_previous=%7B%7D; product_data_storage=%7B%7D; mage-cache-sessid=true; mage-messages=; private_content_version=3ff42898bcbcd465fedd690b41db0810; X-Magento-Vary=9bf9a599123e6402b85cde67144717a08b817412; section_data_ids=%7B%22cart%22%3A1502993675%2C%22customer%22%3A1502993675%2C%22product_data_storage%22%3A1502993675%2C%22product_data_storage%22%3A1502993675%2C%22review%22%3A1502993675%2C%22wishlist%22%3A1502993675%2C%22messages%22%3A1502993751%7D cs_content_length: 0 cs_content_type: application/json dest_content: {"message":"Coupon code is not valid","trace":"#0 [internal function]: Magento\\Quote\\Model\\CouponManagement->set(146, '20twitter')\\n#1 \\var\\www\\html\\magento2\\app\\code\\Magento\\Webapi\\Concall_user_func_array(Array, Array)\\n#2 \\var\\www\\html\\magento2\\app\\code\\Magento\\Webapi\\ConMagento\\Webapi\\Controller\\Rest->processApiRequest()\\n#3 \\var\\www\\html\\magento2\\lib\\internal\\Magento\\FrameworkMagento\\Webapi\\Controller\\Rest->dispatch(Object(Magento\\Framework\\App\\Request\\Http))\\n \\var\\www\\html\\magento2\\lib\\internal\\Magento\\FrameworkMagento\\Webapi\\Controller\\Rest\\Interceptor

Add Panel → New → Statistics Table → title it “Web Path Enumeration” → Shared Time Picker.

Empty tables here are a normal, expected result — it means nothing in your current filter selection crossed the threshold, not that the panel is broken. Set both dropdowns to “all” to maximize your chance of seeing rows the first time you test this.

PART 11 Turn a Detection Into a Real Alert

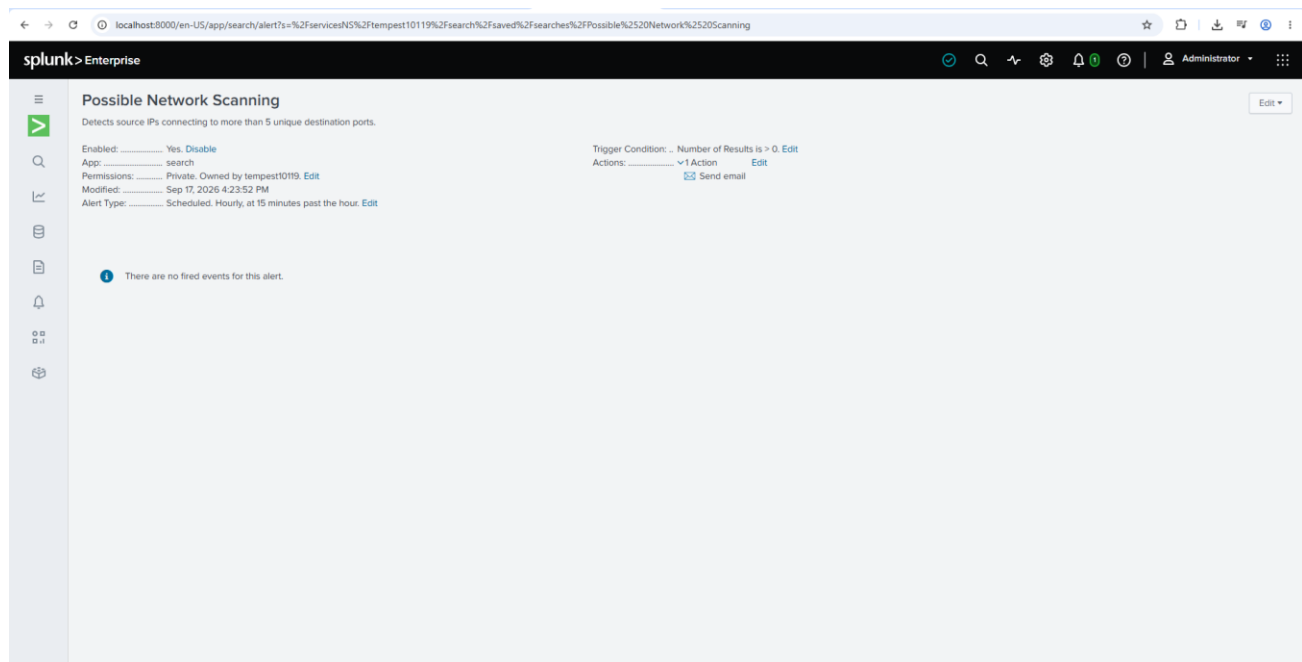
GOAL: Understand the difference between something you look at (a dashboard panel) and something that looks for you (a scheduled alert) — and build one.

A dashboard panel only runs while the page is open in front of you. An alert runs on its own schedule, forever, and notifies you. Same search, different destination. Let's promote the scanning detection into a real, standalone alert.

1. Go to Search & Reporting and paste this search (note: no \$tokens\$ — alerts run on their own and can't reference dashboard inputs):

```
index=botsv2 earliest=-24h
| stats dc(dest_port) as unique_ports count as events by src_ip
| where unique_ports > 5
```

2. Run the search once to confirm it returns results.
3. Click Save As → Alert.
4. Alert type: Scheduled.
5. Set it to run on a schedule, for example every 15 minutes.
6. Trigger Condition: Number of Results — is greater than — 0.
7. Trigger Actions: Add Actions → choose Send Email (or Log Event / Webhook, depending on what's configured in your environment).
8. Click Save.



Notice earliest=-24h here instead of earliest=0. A scheduled alert needs a rolling window that matches how often it runs — otherwise it will re-fire on the exact same old BOTS v2 events every single time it runs, forever. This is the single most common mistake when converting a dashboard search into an alert.

Completion Checklist

Check every box before considering this activity complete.

- ☒ Dashboard titled “BOTS v2 Security Overview” exists and is saved.
- ☒ Total Events panel shows a single large number.
- ☒ Three pie charts (Sourcetype, Host, Hour of Day) all render data.
- ☒ Recent Event Detail table shows readable rows.
- ☒ Sourcetype, Host, and Time Range inputs all appear at the top of the dashboard.
- ☒ A green Submit button is visible next to the inputs.
- ☒ Selecting one specific Sourcetype and clicking Submit visibly changes every panel.
- ☒ Three detection panels (High-Volume Accounts, Possible Network Scanning, Web Path Enumeration) are added and set to Shared Time Picker.
- ☒ One detection has been saved as a real, scheduled Alert with a trigger action configured.
- ☒ You can explain, out loud, the difference between a dashboard panel and an alert without notes.

If Something Isn't Working

Symptom	Fix
A panel shows 0 results	Check the panel's search still has \$st_tok\$/ \$host_tok\$ correctly typed, and its time range is set to Shared Time Picker, not a fixed range.
All panels show 0 results, even Total Events	Check your Time Range input's default is Earliest = 0. BOTS v2 data is from 2017 and will not appear under a default “last 24 hours” range.
No Submit button appears	Re-check Part 7 — the Autorun Dashboard checkbox must be unchecked and the dashboard re-saved.
Changing a dropdown does nothing, even after clicking Submit	That panel's search was never updated with the token in Part 8. Re-open its Edit Search and confirm \$st_tok\$ or \$host_tok\$ actually appears in the query text.
The Time Range input looks greyed out and unclickable	It was added as a Dropdown instead of the dedicated Time input type. Delete it and re-add it choosing Time.